



Introducing: Field-Programmable Gate Arrays (FPGAs)

Ian F.V.G. Hunter, Out standing in his field, Ireland

Abstract

In this paper we introduce a new approach to agricultural management through our innovative system, dubbed “Field Programmable Gate Arrays” or FPGAs. By placing moisture sensors in key locations in crop fields, we assess the suitability of each field for harvesting. Based on these data, the motorized mechanisms in our array of gates will open or close their corresponding gate. L.E.D. displays atop the gates will indicate to farmers the most efficient routes for travelling between fields.

By providing farmers with real-time, no-touch harvest readiness data we can promote sustainable farming, avoid land destruction and streamline food production and with further development, we expect it may solve world hunger.

1 Introduction

While FPGAs have existed in casual use in farming communities for centuries, the academic footprint is littered with bogus publications describing configurable ‘Integrated Circuits’¹. Instead, much farming knowledge has been passed down generation-to-generation, refined by learnings of each harvest. HPC workers (Humans with Powered Contraptions) have opened gates and gate arrays manually for so long, the idea of writing such an obvious idea down was laughable. The term ‘FPGA’ has been used since before records begun, to describe the aspirational future of having a hands-free gate opening system.



Figure 1: This is a gate. It is **Closed**. For the sheep to be able to pass it, it would need to be **Open**. Photo Credit: Rosa Stone

Because of this sniping of terms and obfuscation of the few records that do exist, it is prudent that we briefly recap the existing literature:

¹Shortened begrudgingly to in these papers as ICs, which is another overloaded term and should refer to intercropping, the process of growing multiple crops in the same field — An actual parallel system for heterogeneous work.

- Some gates can be opened if you push them.
- Some gates can be opened if you pull them.
- Some gates can be opened by either pushing or pulling.
- Some gates cannot be opened. This state is known as ‘Locked’
- If a ‘Locked’ gate becomes opened at a future point, it becomes ‘Unlocked’
- Technology exists to open gates automatically [5].

But, there is, as-of-yet, no true implementation of a FPGA. Hence, in the following sections, we will detail our approach to making this long-wanted dream a reality.

2 Design

2.1 Basic Overview of an FPGA

In Figure 2, a hardware diagram of our FPGA can be seen. For each acre of field, we place one sensor in its center. Our initial prototype uses a moisture sensor, but in the future we expect that FPGAs will have more sensors for detecting wildlife, temperature, pH levels, and nutrient levels such as nitrogen. We’ll probably put some Artificial Intelligence in there too because that seems to get you funding.

These sensors are attached to an Arduino microcontroller[1] stored safely in the hollow of each gate pole. A cap can be put on these hollows to protect the microcontroller from the elements. This microcontroller operates the triggering of the gate open/close mechanism and a waterproof L.E.D. display.

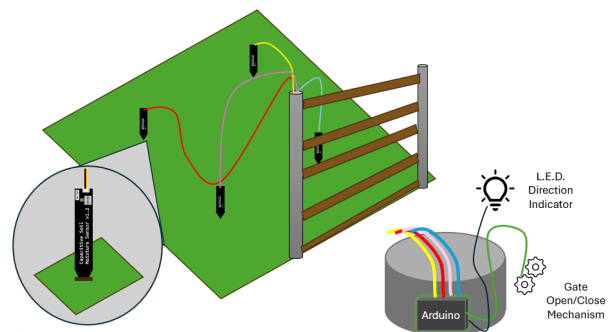
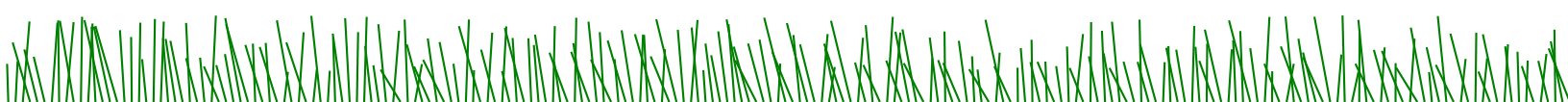


Figure 2: Hardware Diagram. Sensors are placed in soil and wired to a central pole. This diagram depicts a four-acre field with one sensor per acre.

If the microcontroller calculates an average moisture level above a farmer-defined threshold, it will close the gate and display a red ‘X’ on the L.E.D. screen. It will also open the gate if it falls below the threshold. Due to the volatility of these sensors, we implement a fuzzy logic system² to ensure that gates do not swing open and

²Again, another term stolen from the farming industry, though





closed wildly with fluctuating moisture readings. Figure 3 depicts this system in use on a hypothetical farm.

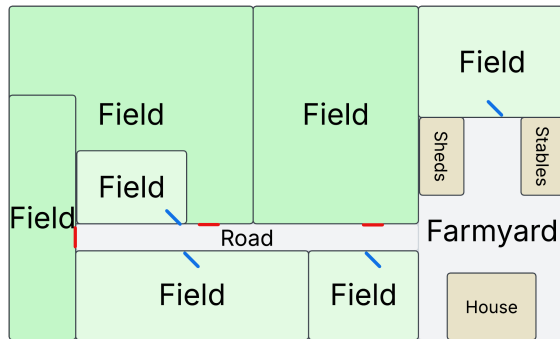


Figure 3: An example of an FPGA in use. Most fields have their gates open. Moist fields have their gates closed.

2.2 The Travelling Farmer Problem

Initial tests of this system were very successful. However, one consistent problem kept occurring — While the gates worked individually, they had no understanding of the other nodes in the system. Farmers would need to drive up to every gate to see if it was open. This was resolved with the introduction of the L.E.D. display which was not present in alpha prototypes. Each gate has its own hexadecimal display which signals a status or instruction to the farmer as he passes each gate.

- **0xCOFFEE** - All harvest-ready farms have been visited. The farmer can leave for lunch, or other activities.
- **0xDODEED** - Instructs the farmer to drive through this gate and harvest this field.
- **0xBADSEED** - This field is not ready for harvest.
- **0xDADBOD** - A test code I accidentally left in.

By integrating a omni-directional field-to-field transportation optimal route algorithm, advanced farmers can choose for their signs to display an optional directional arrow instead of the textual messages. This is particularly useful with complex field patterns and pathways. While an expert will create an initial hardware mapping for the farmer, it may be finely tuned if wished.

For farmers who have difficulty with the messages, arrows or configurations, there is also a simple binary mode where the L.E.D. screen simply turns all pixels ON or OFF depending on whether the field is ready or not.

3 Conclusion

In this paper we have introduced FPGAs as a new transformative agricultural management technology. Our system allows farmers to make informed harvesting decisions, optimizes petroleum usage for a more eco-friendly future, reduces the labour cost of manual gate-openers and gate-closers and brings forth a system desired for generations, into reality.

they are now modernly called Sheepdogs

3.1 Future Work

Ternary FPGAS For farmers who like their own judgement, we would be interested to see an FPGA which provides DONT_CARE or DONT_KNOW statuses for cases which are inconclusive or a gamble.

Deployment at scale Due to funding limitations, we only have four gates and two fields in which to test our solution. It would be interesting to examine FPGAs at scale, in a large farm. Please buy me a farm.

The Delicate Nature of FPGAs One issue with this system is that the delicate wires can be driven over by a tractor and destroyed, or ripped out by machinery or animals. It would be interesting to see a distributed wireless system with sensor nodes either dug deeper than ploughs can reach, or a drone system that can remove the sensors as a tractor enters a field.

ASICs Although FPGAs are optimized for crop production, we believe that there are other technologies that are at a point now where serious breakthroughs can be made. A significant example is that of ASICs (Animals Scrapping Irregular Crops) where FPGAs can be used to control the movement of cattle and goats so that they are directed by the gates to eat the lower quality crops in a farmer's field, while the premium crops are saved for human consumption.

3.2 Disclaimer

It has come to our attention that this research is not novel. [9], [7], [4], [2], [6], [3], [8]. Please disregard this fact and find \$20 enclosed for your trouble.

3.3 Acknowledgements

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